TATA

TATA ELXSI

TELIPORT

Script Your Future

Season 4

THINK. BUILD. CREATE. WIN

CASE STUDY 4

TELIPORT

Script Your Future

Season 4



End-to-End Labeling Automation Management for Medical Devices

Theme: Intelligent Medical Device Labeling Automation Platform

Design innovative, scalable, and compliant solutions that transform the end-to-end medical device labeling lifecycle by leveraging Artificial Intelligence (AI), Machine Learning (ML), Generative AI, Robotics Process Automation (RPA), Computer Vision, Smart Workflows, and Cloud Technologies.

The challenge encourages participants to rethink traditional labeling operations and create future-ready solutions that accelerate regulatory compliance, reduce human effort, minimize errors, and enhance patient safety.

Key Takeaways: Rethink traditional medical device labeling operations and create future-ready solutions that accelerate regulatory compliance, reduce human effort and labeling design & implementation cycle time, minimize errors, and enhance patient safety.

Background

Medical devices are sold in highly regulated markets where the information supplied with the device is not simply packaging artwork. The label, packaging information, markings, symbols, instructions for use (IFU), warnings, precautions, Unique Device Identification (UDI) and related information communicate critical information about the device and its safe use.

Medical device labeling is one of the most critical regulatory and patient-facing components of a medical device lifecycle.

A single label may contain (Not limited to):

- Product identification
- Unique Device Identification (UDI)
- Regulatory symbols
- Country-specific requirements
- Multi-language content
- Safety warnings
- Sterilization information
- Traceability data
- Manufacturing details

Today many organizations still rely heavily on (Not limited to):

- Labeling Specialized Resources (e.g. people, tools, change management system etc.)
- Traditional Labeling Design practices & standards operating procedures (SOPs)
- Manual content updates
- Spreadsheet-based tracking and monitoring
- Multiple approval cycles and delayed approvals
- Human verification and validation
- Repetitive artwork generation
- Fragmented systems (e.g. multiple PLMs, SAP etc.)

TELIPORT

Script Your Future

Season 4



As products expand globally, maintaining label compliance across multiple markets becomes increasingly complex. The label design, regulatory approval and implementation cycle time sometime becomes roadblock and impact the product launches and resume production for any labeling change.
The medical device industry needs innovative solutions that combine automation, intelligence, compliance, scalability and reduce design & implementation cycle time.

Key Takeaways:

- **Automation** (What manual activities can be eliminated, how can repetitive tasks be automated etc)
- **AI Driven** (AI reviews labels, GenAI generate compliant content etc)
- **Compliance** (meet all the regulatory requirements be embedded into systems)
- **Innovation** (labeling become "self-validating" and labels predict compliance risks before release)
- **Digital Transformation** (Can Industry 4.0 concepts be applied?)
- **Patient Safety** (How can labeling errors be prevented before reaching the market)

Stakeholders Involved:

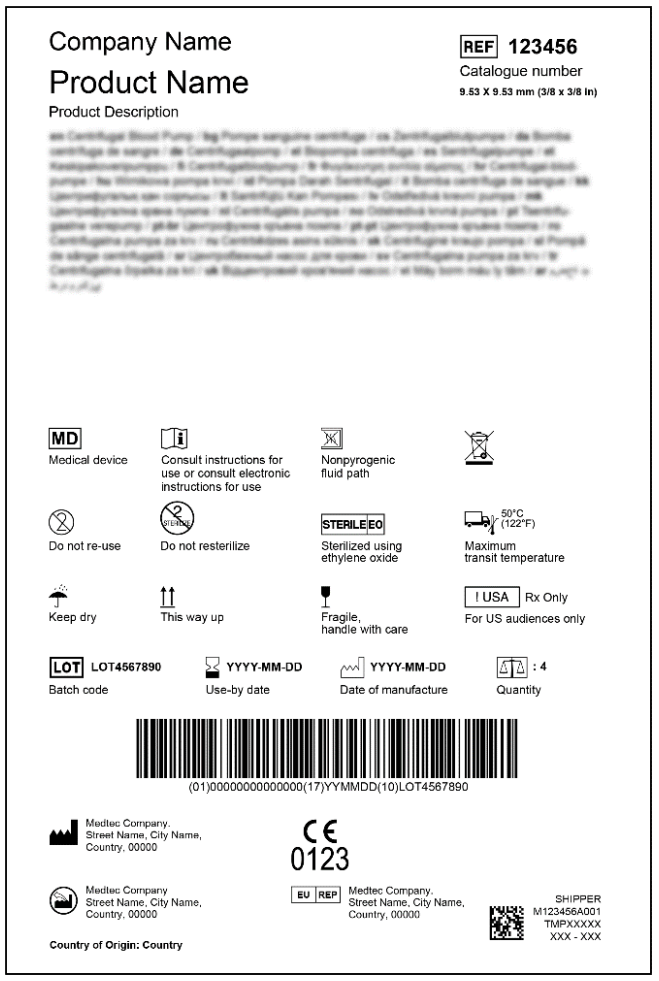
- Regulatory Affairs for Regulatory compliance management and Change impact assessment
- Product Owner(s)
- Labeling Engineers for Label creation and maintenance
- Quality Assurance for Verification and validation
- Manufacturing for Print readiness and production release
- Supply Chain for Product traceability
- Clinical Teams for Safety and usage information
- IT & Digital Transformation Teams for Platform implementation

Regulatory Authorities (Not limited to): EU MDR, US FDA, Global Submission

Business Process in Scope:

Traditional Approach:

Product / Design Information → Product Understanding → Identify Applicable Markets → Regulatory Requirement Assessment → Gap Assessment → Identify Labeling Gaps → Gap Addressal → Label Content Development → Redline / Artwork Modification → Stakeholder Review → Regulatory / Quality Approval → Label Revision → Change Order Creation → Document / Configuration Control → Final Quality Check → Label Release → Manufacturing → Production Implementation



TATA ELXSI

TELIPORT

Script Your Future

Season 4



Expectation: New Approach using AI with artifacts or supported data

Problem (Expanded – Real-World Context)

Current Industry Challenges:

Imagine a global manufacturer selling: Cardiac stents, Surgical instruments, Diagnostic devices, Orthopaedic implants across various geographies like United States, Europe, Japan, China, Middle East then, a single product change may require updates to:

- 30+ labels
- 20+ languages
- Multiple packaging levels
- Regional regulatory requirements

The Common Pain Points which are leading to human errors.

1. Manual Data Entry
2. Information is copied between:
   - ERP systems
   - PLM systems
   - Labelling software
   - Regulatory databases

Regulatory Complexity:

1. Requirements continuously evolve. Examples:
   - UDI updates
   - New symbols
   - MDR changes
   - Country-specific requirements
2. Review process creating delays. Multiple reviewers manually compare:
   - Old versions
   - New versions
   - Regulatory documents
3. Translation Challenges- Translation inconsistencies may introduce:
   - Safety risks
   - Regulatory observations
   - Product recalls
   - Artwork Verification
4. Visual inspection remains heavily dependent on humans
5. Thousands of parts/devices create enormous operational burden

Systemic Impact Summary:

Failure in labeling processes may result in:

- Business Impact
- Market launch delays or resume production for design change cases
- Revenue loss
- Increased operational costs

TELIPORT

Script Your Future

Season 4



Regulatory Impact:

- EU MDR/FDA observations
- Warning letters
- Non-conformities
- Product recalls/holds

Quality Impact:

- Labeling defects
- Version mismatches

Patient Safety Impact:

- Incorrect product usage
- Adverse events
- Misidentification

Sustainability Impact

- Rework
- Scrap generation
- Inventory waste

Proposed Solutions should explicitly address:

1. **Time-Consuming Manual Process-** Experts manually review specifications, regulations, labels, gaps, redlines, revisions and change documentation. This creates high cycle time, repetitive work, expert dependency and scalability challenges.
2. **Frequent Regulatory Updates -** Teams manually monitor new regulations, revised standards, new/revised symbols, UDI requirements and market-specific requirements. The key risk is missing affected labels.
3. **Symbol Management -** Experts manually determine applicable symbols, correct versions, placement, associated text and market applicability. Errors can cause compliance issues, rework and production delays.
4. **Multiple Sources of Truth -** Relevant information may reside in PLM, ERP, requirements documents, regulatory databases, specifications, spreadsheets, artwork files, previous labels and quality systems.
5. **Expertise Dependency** - Labeling requires product, engineering, regulatory, symbols, UDI, quality, artwork and manufacturing knowledge, creating bottlenecks and training challenges.
6. **Manual Redlining -** Experts manually compare current and required labels and create redlines. Computer vision and document intelligence could automate difference detection.
7. **Repetitive Review -** The same checks—product number, manufacturer, symbols, UDI, dates, warnings, language and formatting—are repeatedly performed and are candidates for rules-based automation.
8. **Traceability -** Teams need to answer why each label element exists and trace it to product data, a requirement, source document, approval and change order.
9. **Variant Explosion** - Multiple products, configurations, markets and languages can create thousands of potential label combinations.
10. **Human Error -** Manual copy/paste, outdated symbols, incorrect identifiers, missing information, inconsistent terminology, wrong revisions and wrong market information can introduce risk.

TELIPORT

Script Your Future

Season 4



Evaluation Criteria

- Understanding of labeling problem
- Regulatory/compliance awareness
- Innovation ideas considering Label Design using AI
- AI/automation platform
- End-to-end process coverage
- Accuracy & validation approach
- Traceability & auditability

Recommended Steps

- Discover & Identify the Manual activities, Process bottlenecks, Error-prone tasks to create Process Map
- Analyse & Perform Root cause analysis/Failure Mode Analysis/Value Stream Mapping to create Automation Opportunity Matrix
- Innovate to Identify technologies applicability such as Artificial Intelligence, Generative AI, Machine Learning, Computer Vision, RPA, Cloud Architecture etc.
- Design & development of labels using AI
- Design & create system architecture, Automation workflow
- Measure and evaluate Time reduction, Error reduction, Compliance improvement, Cost savings

Potential Approaches

1. **GenAI Label Authoring Assistant -** Capabilities:

- Generate compliant label text
- Regulatory language suggestions
- Auto-update safety statements

2. **AI Compliance Checker -**Uploads:

- Label PDF
- Artwork file

AI automatically verifies:

- Missing information
- Symbol usage
- Regulatory compliance

3. Smart Change Impact Engine- When a product parameter changes: AI identifies

- Impacted labels
- Impacted markets
- Required actions

4. Computer Vision-Based Artwork Validation -System compares:

- Approved artwork
- Revised artwork

TELIPORT

Script Your Future

Season 4



Automatically identifies:

- Text changes
- Barcode changes
- Layout deviations

5. Autonomous Regulatory Intelligence Bot - Continuously monitors:

- FDA updates
- EU MDR updates
- Global regulations

Provides proactive alerts.

6. Smart Translation Validation Platform - Uses AI to:

- Validate translations
- Detect inconsistencies
- Suggest harmonized terminology

7. Voice/Prompt to Label Creation- Users describe product changes verbally/written. AI automatically:

- Creates draft labels
- Generates change requests
- Initiates approval workflows

8. Predictive Label Quality Analytics - System predicts below before release:

- Potential errors
- High-risk products
- Compliance failures

9. Augmented Reality Label Verification - Using AR glasses/mobile devices:

- Real-time label inspection
- Visual discrepancy alerts
- Guided quality checks

10. Agentic AI Labeling Ecosystem (Future Vision) - Multiple AI agents collaborate to:

- Write labels (Design & Development of labels)
- Validate compliance
- Review changes
- Execute workflows
- Generate audit records with minimal human intervention.

TELIPORT

Script Your Future

Season 4



References & Study Material

1. FDA – General Device Labeling Requirements: https://www.fda.gov/medical-devices/device-labeling/general-device-labeling-requirements

2. EU MDR – Regulation (EU) 2017/745: https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A02017R0745-20230320

3. ISO 20417:2026 – Information supplied by the manufacturer: https://www.iso.org/standard/88714.html

4. ISO 15223-1 – Symbols used with medical devices: https://www.iso.org/standard/77326.html

5. IMDRF N52 – Principles of Labelling: https://www.imdrf.org/documents/principles-labelling-medical-devices-and-ivd-medical-devices

6. European Commission – Unique Device Identifier (UDI): https://health.ec.europa.eu/medical-devices-topics-interest/unique-device-identifier-udi_en

7. FDA – Unique Device Identification System: Form and Content of UDI: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/unique-device-identification-system-form-and-content-unique-device-identifier-udi